

✓ Exercise 2: Predict Body Weight from Height and Age

Conduct Linear Regression on a model that predicts the body weight of a person. The dependent variable is body weight, while height and age are independent variables.

```
#CH.SC.U4CSE24163 HARISH.R

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

✓ Importing Dataset

```
df = pd.read_csv('../Data/linregnp3_dataset.csv')
print(f"Shape: {df.shape}")
df.head()
```

Shape: (40, 3)

	Height	Age	Weight
0	1.80	35	65.0
1	1.68	39	65.6
2	1.82	25	69.2
3	1.70	60	73.3
4	1.87	27	73.0

✓ Assigning Dependent and Independent variables

```
x = df[['Height', 'Age']]
y = df['Weight']
```

✓ Splitting the dataset into Training and Testing Dataset

```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random
```

```
print(f"Training set size: {x_train.shape[0]}")  
print(f"Testing set size: {x_test.shape[0]}")
```

```
Training set size: 32  
Testing set size: 8
```

✓ Fitting the Model (Linear Regression)

```
mdl = LinearRegression()  
mdl.fit(x_train, y_train)  
print(f"Intercept: {mdl.intercept_:.2f}")  
print(f"Coefficients: {mdl.coef_}")
```

```
Intercept: 9.07  
Coefficients: [30.03553051  0.16990121]
```

✓ Evaluation Metrics

```
y_pred = mdl.predict(x_test)  
mse = mean_squared_error(y_test, y_pred)  
r2 = r2_score(y_test, y_pred)  
print("Eval Metrics")  
print(f"MSE: {mse:.2f}")  
print(f"R2: {r2:.2f}")
```

```
Eval Metrics  
MSE: 3.55  
R2: 0.45
```

